Complex Analysis Qualifying Exam Problems

Pic S. Infiniti (Second Edition)

Summer 2026-2027

Contents

1 August 2018 **2**
1.1 Part I 2
1.2 Part II 10

2 August 2020 **17**
2.1 Part I 17
2.2 Part II 22

3 August 2024 **25**
3.1 Part I 25
3.2 Part II 26

4 January 2025 **29**
4.1 Part I 29
4.2 Part II 32

August 2018

Part I

1. *Find a conformal map f from the vertical strip*

$$\Omega = \{z \in \mathbb{C} : 3 < \mathrm{Re}(z) < 11\}$$

onto the open unit disk $\mathbb{D}$.

Solution. The width of the strip is $11 - 3 = 8$. First map the strip to a horizontal strip by setting

$$w = \frac{\pi}{8}(z-3).$$

Since $3 < \mathrm{Re}(z) < 11$, we have

$$0 < \mathrm{Re}(w) < \pi.$$

Now define

$$\omega = e^{iw}.$$

Because

$$\arg(\omega) = \mathrm{Re}(w),$$

the map $\omega = e^{iw}$ sends the strip onto the upper half-plane

$$\{\omega \in \mathbb{C} : \mathrm{Im}(\omega) > 0\}.$$

Finally, the map

$$\phi(\omega) = \frac{\omega - i}{\omega + i}$$

maps the upper half-plane conformally onto the unit disk $\mathbb{D}$. Therefore a conformal map from Ω onto $\mathbb{D}$ is

$$\boxed{f(z) = \frac{e^{i\pi(z-3)/8} - i}{e^{i\pi(z-3)/8} + i}}.$$

∎

2. *Suppose that F and G are holomorphic functions on the right half plane*

$$H = \{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}.$$

Suppose also that F and G have the same zeros with the same multiplicities at each zero. Prove that there is a holomorphic function h on H such that

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$.

Solution. If $F \equiv G \equiv 0$, then we may take $h \equiv 0$. Thus assume that F and G are not identically zero.

Define

$$q(z) = \frac{F(z)}{G(z)}$$

at points where $G(z) \neq 0$. We claim that q extends to a holomorphic function on all of H.

Let $a \in H$ be a zero of G. Since F and G have the same zeros with the same multiplicities, there is some $m \geq 1$ such that locally

$$F(z) = (z-a)^m \phi(z), \qquad G(z) = (z-a)^m \psi(z),$$

where ϕ and ψ are holomorphic near a, with

$$\phi(a) \neq 0, \qquad \psi(a) \neq 0.$$

Hence near a,

$$q(z) = \frac{F(z)}{G(z)} = \frac{\phi(z)}{\psi(z)}.$$

So the singularity of q at a is removable. Therefore q extends holomorphically to all of H.

Moreover, the extension of q has no zeros. Indeed, away from the zeros of G, this is clear since F and G have the same zeros. At a common zero a, we have

$$q(a) = \frac{\phi(a)}{\psi(a)} \neq 0.$$

Now $H = \{z \in \mathbb{C} : \mathrm{Re}(z) > 0\}$ is convex, hence simply connected. Since q is holomorphic and never zero on H, by Corollary 6.17, Chapter IV, there exists a holomorphic function h on H such that

$$q(z) = e^{h(z)}.$$

Therefore, for all $z \in H$,

$$F(z) = q(z)G(z) = e^{h(z)}G(z).$$

Thus there is a holomorphic function h on H satisfying

$$F(z) = e^{h(z)}G(z)$$

for all $z \in H$. ∎

3. *Suppose f is a holomorphic function on the open unit disc*

$$D = \{z : |z| < 1\}.$$

Prove that there is a sequence $(z_n)_{n=1}^{\infty}$ in D such that $|z_n| \to 1$ and the set

$$\{f(z_n) : n \geq 1\}$$

is bounded.

Solution (I). If $f \equiv 0$, then take $z_n = 1 - \frac{1}{n}$. Then $|z_n| \to 1$ and $\{f(z_n) : n \geq 1\} = \{0\}$, which is bounded.

Now assume $f \not\equiv 0$. Suppose, toward a contradiction, that no such sequence exists. Then for every $M > 0$, there is $0 < \rho < 1$ such that

$$|f(z)| > M$$

whenever $\rho < |z| < 1$. Indeed, otherwise we could find $z_n \in D$ with

$$|z_n| > 1 - \frac{1}{n} \qquad \text{and} \qquad |f(z_n)| \leq M,$$

which would give $|z_n| \to 1$ and $\{f(z_n)\}$ bounded.

Thus

$$|f(z)| \to \infty \qquad \text{as } |z| \to 1.$$

In particular, for some $0 < R < 1$,

$$|f(z)| > 1$$

whenever $R < |z| < 1$. Hence all zeros of f lie in the compact set $\{|z| \leq R\}$. Since $f \not\equiv 0$, it has only finitely many zeros there. Let those zeros be $\alpha_1, \ldots, \alpha_N$, with multiplicities $m_1, \ldots, m_N$, and define

$$P(z) = \prod_{j=1}^{N} (z - \alpha_j)^{m_j}.$$

If f has no zeros, take $P \equiv 1$.

Then

$$F(z) = \frac{f(z)}{P(z)}$$

extends to a holomorphic function on D, and F has no zeros in D. Therefore

$$g(z) = \frac{1}{F(z)}$$

is holomorphic on D.

Since P is bounded on D, and $|f(z)| \to \infty$ as $|z| \to 1$, we get

$$|F(z)| \to \infty \qquad \text{as } |z| \to 1.$$

Thus

$$g(z) \to 0 \qquad \text{as } |z| \to 1.$$

Fix $z_0 \in D$. Given $\varepsilon > 0$, choose $\rho < 1$ such that

$$|g(z)| < \varepsilon$$

whenever $\rho < |z| < 1$. Choose r such that

$$\max\{|z_0|, \rho\} < r < 1.$$

By the maximum modulus principle applied on $\{|z| \le r\}$,

$$|g(z_0)| \le \max_{|z|=r} |g(z)| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $g(z_0) = 0$. Hence $g \equiv 0$ on D, which is impossible because $g = 1/F$.

Therefore our supposition was false. Hence there exists a sequence $(z_n) \subset D$ such that

$$|z_n| \to 1$$

and

$$\{f(z_n) : n \ge 1\}$$

is bounded. ■

Solution (II). Suppose, toward a contradiction, that no such sequence exists. Then for every $M > 0$, there exists $0 < \rho < 1$ such that

$$|f(z)| > M$$

whenever $\rho < |z| < 1$. Otherwise, for some $M > 0$, we could choose $z_n \in D$ with

$$|z_n| > 1 - \frac{1}{n} \qquad \text{and} \qquad |f(z_n)| \leq M,$$

which would give the desired sequence.

Thus

$$|f(z)| \to \infty \qquad \text{as } |z| \to 1.$$

In particular, there exists $0 < R < 1$ such that

$$|f(z)| > 1$$

whenever $R < |z| < 1$. Hence f has no zeros in the annulus

$$A = \{z : R < |z| < 1\}.$$

Therefore

$$g(z) = \frac{1}{f(z)}$$

is holomorphic on A. Also,

$$g(z) \to 0 \qquad \text{as } |z| \to 1.$$

Now expand g in its Laurent series on the annulus A:

$$g(z) = \sum_{n=-\infty}^{\infty} c_n z^n.$$

For any r with $R < r < 1$,

$$c_n = \frac{1}{2\pi i} \int_{|\omega|=r} \frac{g(\omega)}{\omega^{n+1}}\, d\omega.$$

Hence

$$|c_n| \leq \max_{|\omega|=r} |g(\omega)|\, r^{-n}.$$

Letting $r \to 1^-$, we have

$$\max_{|\omega|=r} |g(\omega)| \to 0,$$

and $r^{-n} \to 1$. Therefore

$$c_n = 0$$

for every $n \in \mathbb{Z}$. Hence

$$g \equiv 0$$

on A, which is impossible since $g = 1/f$.

Therefore our supposition was false. Thus there exists a sequence $(z_n) \subset D$ such that

$$|z_n| \to 1$$

and

$$\{f(z_n) : n \geq 1\}$$

is bounded. ■

4. *Determine, with proof, all entire functions f such that for all $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and all $z \in \mathbb{C}$ the equation*

$$f(\lambda z) = \lambda f(z)$$

holds.

Solution (I). We claim that the only such functions are

$$f(z) = cz$$

for some constant $c \in \mathbb{C}$.

Let f be entire. Then f has a power series expansion on all of $\mathbb{C}$:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

For $|\lambda| = 1$, we have

$$f(\lambda z) = \sum_{n=0}^{\infty} a_n (\lambda z)^n = \sum_{n=0}^{\infty} a_n \lambda^n z^n.$$

By assumption,

$$f(\lambda z) = \lambda f(z) = \lambda \sum_{n=0}^{\infty} a_n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

Therefore, for every $|\lambda| = 1$,

$$\sum_{n=0}^{\infty} a_n \lambda^n z^n = \sum_{n=0}^{\infty} \lambda a_n z^n.$$

By uniqueness of power series coefficients,

$$a_n \lambda^n = \lambda a_n$$

for every $n \geq 0$ and every $|\lambda| = 1$. Hence

$$a_n(\lambda^n - \lambda) = 0.$$

If $n = 1$, this condition gives no restriction on a_1. But if $n \neq 1$, we can choose $|\lambda| = 1$ such that

$$\lambda^n \neq \lambda.$$

Thus $a_n = 0$ for every $n \neq 1$. Therefore

$$f(z) = a_1 z.$$

Conversely, every function of the form $f(z) = cz$ satisfies

$$f(\lambda z) = c\lambda z = \lambda c z = \lambda f(z).$$

Hence the entire functions satisfying the condition are exactly

$$\boxed{f(z) = cz, \qquad c \in \mathbb{C}.}$$

■

Solution (II). First put $z = 0$. Then

$$f(0) = \lambda f(0)$$

for every $|\lambda| = 1$. Choosing $\lambda \neq 1$, we get

$$f(0) = 0.$$

Hence

$$g(z) = \frac{f(z)}{z}$$

is holomorphic on $\mathbb{C} \setminus \{0\}$, and the singularity at 0 is removable. Define

$$g(0) = f'(0).$$

Then g is entire.

For $z \neq 0$ and $|\lambda| = 1$,

$$g(\lambda z) = \frac{f(\lambda z)}{\lambda z} = \frac{\lambda f(z)}{\lambda z} = \frac{f(z)}{z} = g(z).$$

Thus, by continuity,

$$g(\lambda z) = g(z)$$

for all $z \in \mathbb{C}$ and all $|\lambda| = 1$.

In particular, if $|\zeta| = 1$, then $\zeta = \lambda \cdot 1$ for some $|\lambda| = 1$, so

$$g(\zeta) = g(1).$$

Thus g is constant on the unit circle.

Write the Taylor expansion of g at 0:

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

By Cauchy's integral formula for Taylor coefficients,

$$a_n = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{g(\zeta)}{\zeta^{n+1}}\, d\zeta.$$

For $n \geq 1$, since $g(\zeta) = g(1)$ on $|\zeta| = 1$, we get

$$a_n = \frac{g(1)}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta^{n+1}}\, d\zeta = 0.$$

Therefore all Taylor coefficients of g except possibly a_0 vanish. Hence g is constant on $\mathbb{C}$. Say

$$g(z) = c.$$

Then

$$f(z) = zg(z) = cz.$$

Conversely, if $f(z) = cz$, then

$$f(\lambda z) = c\lambda z = \lambda cz = \lambda f(z).$$

Therefore the desired functions are exactly

$$\boxed{f(z) = cz, \qquad c \in \mathbb{C}.}$$

■

5. *Let f and g be entire functions satisfying $f(0) = g(0) \neq 0$ and*

$$|f(z)| \leq |g(z)| \qquad \forall z \in \mathbb{C}.$$

Show that $f = g$.

Solution. Since $g(0) = f(0) \neq 0$, there exists $r > 0$ such that

$$g(z) \neq 0$$

for all $z \in B(0, r)$. Define
$$h(z) = \frac{f(z)}{g(z)}$$
on $B(0, r)$. Then h is holomorphic on $B(0, r)$. Also, by hypothesis,
$$|h(z)| = \frac{|f(z)|}{|g(z)|} \leq 1$$
for all $z \in B(0, r)$. Moreover,
$$h(0) = \frac{f(0)}{g(0)} = 1.$$
Thus
$$|h(0)| = 1.$$
So h attains its maximum modulus at the interior point 0. By the maximum modulus principle, h is constant on $B(0, r)$. Since $h(0) = 1$, we get
$$h \equiv 1$$
on $B(0, r)$. Therefore
$$f(z) = g(z)$$
for all $z \in B(0, r)$.

Since f and g are entire and agree on a nonempty open set, the identity theorem implies that
$$f = g$$
on all of $\mathbb{C}$. ■

Part II

6. *Determine all entire functions that satisfy the functional equation*
$$f(3z) = (1 - 3z)f(z).$$
Express the answer in terms of an infinite product.

Solution. Replacing z by $z/3$ in the functional equation gives
$$f(z) = (1 - z)f\left(\frac{z}{3}\right).$$

Iterating this identity n times, we obtain

$$f(z) = \left(\prod_{k=0}^{n-1} \left(1 - \frac{z}{3^k}\right) \right) f\left(\frac{z}{3^n}\right).$$

Since f is entire, it is continuous, and hence

$$\lim_{n\to\infty} f\left(\frac{z}{3^n}\right) = f(0).$$

Now define

$$P(z) = \prod_{k=0}^{\infty} \left(1 - \frac{z}{3^k}\right).$$

This product converges uniformly on compact subsets of $\mathbb{C}$. Indeed, for $|z| \leq R$,

$$\sum_{k=0}^{\infty} \left|\frac{z}{3^k}\right| \leq R \sum_{k=0}^{\infty} \frac{1}{3^k} = \frac{3R}{2} < \infty.$$

Therefore, P is entire. Letting $n \to \infty$ in the iterated identity gives

$$f(z) = f(0) \prod_{k=0}^{\infty} \left(1 - \frac{z}{3^k}\right).$$

Thus every entire solution is of the form

$$f(z) = C \prod_{k=0}^{\infty} \left(1 - \frac{z}{3^k}\right), \qquad C \in \mathbb{C}.$$

Conversely, suppose

$$f(z) = C \prod_{k=0}^{\infty} \left(1 - \frac{z}{3^k}\right).$$

Then

$$\begin{aligned} f(3z) &= C \prod_{k=0}^{\infty} \left(1 - \frac{3z}{3^k}\right) \\ &= C(1-3z) \prod_{k=1}^{\infty} \left(1 - \frac{z}{3^{k-1}}\right) \\ &= (1-3z) f(z). \end{aligned}$$

Hence all entire solutions are

$$f(z) = C \prod_{k=0}^{\infty} \left(1 - \frac{z}{3^k}\right), \qquad C \in \mathbb{C}.$$

■

7. *Let $F(z)$ be holomorphic on a disc $D(z_0, r)$ so that*

$$F(z_0) = F'(z_0) = 0 \qquad \textit{and} \qquad F''(z_0) \neq 0.$$

Show that there are two curves Γ_1 and Γ_2 passing through z_0 that are orthogonal at z_0 and so that the function F when restricted to Γ_1 is real-valued and has a minimum at z_0, and when restricted to Γ_2 it is also real-valued and has a maximum at z_0.

Solution. Since $F(z_0) = F'(z_0) = 0$ and $F''(z_0) \neq 0$, the function F has a zero of order exactly two at z_0. Hence, in a sufficiently small neighborhood of z_0, we may write

$$F(z) = (z - z_0)^2 g(z),$$

where g is holomorphic and

$$g(z_0) = \frac{F''(z_0)}{2} \neq 0.$$

After shrinking the neighborhood if necessary, g has no zeros. Therefore, g admits a holomorphic square root h, so that

$$h(z)^2 = g(z).$$

Define

$$\phi(z) = (z - z_0)h(z).$$

Then

$$F(z) = \phi(z)^2.$$

Moreover,

$$\phi(z_0) = 0 \qquad \text{and} \qquad \phi'(z_0) = h(z_0) \neq 0.$$

Thus, by the holomorphic inverse function theorem, ϕ is biholomorphic from some neighborhood U of z_0 onto a neighborhood V of 0. Let

$$\psi = \phi^{-1}.$$

Choose $\varepsilon > 0$ sufficiently small so that

$$(-\varepsilon, \varepsilon) \cup i(-\varepsilon, \varepsilon) \subseteq V,$$

and define the curves

$$\Gamma_1 = \{\psi(t) : -\varepsilon < t < \varepsilon\}$$

and

$$\Gamma_2 = \{\psi(it) : -\varepsilon < t < \varepsilon\}.$$

Both curves pass through z_0, since $\psi(0) = z_0$.

If $z = \psi(t) \in \Gamma_1$, then

$$F(z) = \phi(z)^2 = t^2.$$

Thus F is real-valued on Γ_1 and has a strict minimum at z_0.

Similarly, if $z = \psi(it) \in \Gamma_2$, then

$$F(z) = \phi(z)^2 = (it)^2 = -t^2.$$

Thus F is real-valued on Γ_2 and has a strict maximum at z_0.

Finally, parametrizing the curves by

$$\gamma_1(t) = \psi(t) \qquad \text{and} \qquad \gamma_2(t) = \psi(it),$$

we obtain

$$\gamma_1'(0) = \psi'(0) \qquad \text{and} \qquad \gamma_2'(0) = i\psi'(0).$$

Multiplication by i rotates a vector through an angle of $\pi/2$. Therefore, the tangent vectors to Γ_1 and Γ_2 at z_0 are orthogonal, and hence the two curves are orthogonal at z_0. ■

8. *Prove that if f is a non-constant bounded holomorphic function defined on the half plane*

$$\{z : \operatorname{Im} z < 1\},$$

then there is some point on the real axis where the value of f is not real.

Solution. Suppose, for contradiction, that

$$f(x) \in \mathbb{R} \qquad \text{for every } x \in \mathbb{R}.$$

Define

$$g(z) = \overline{f(\overline{z})}.$$

Since f is holomorphic on

$$\{z \in \mathbb{C} : \operatorname{Im} z < 1\},$$

the function g is holomorphic on

$$\{z \in \mathbb{C} : \operatorname{Im} z > -1\}.$$

Both f and g are holomorphic on the strip

$$S = \{z \in \mathbb{C} : -1 < \operatorname{Im} z < 1\}.$$

For every $x \in \mathbb{R}$, we have

$$g(x) = \overline{f(x)} = f(x),$$

because $f(x)$ is real. Since the real axis has accumulation points in S, the identity theorem implies that

$$g(z) = f(z) \qquad \text{for every } z \in S.$$

Therefore, the function

$$F(z) = \begin{cases} f(z), & \operatorname{Im} z < 1, \\ \overline{f(\overline{z})}, & \operatorname{Im} z > -1 \end{cases}$$

is well-defined and entire.

Since f is bounded, there exists $M > 0$ such that

$$|f(z)| \leq M \qquad \text{whenever } \operatorname{Im} z < 1.$$

If $\operatorname{Im} z > -1$, then

$$\operatorname{Im}(\overline{z}) < 1,$$

so

$$|F(z)| = \left|\overline{f(\overline{z})}\right| = |f(\overline{z})| \leq M.$$

Thus F is a bounded entire function. By Liouville's theorem, F is constant. Hence f is constant, contradicting the hypothesis. Therefore, there exists some $x \in \mathbb{R}$ such that

$$f(x) \notin \mathbb{R}.$$

■

9. *Show that there exists a sequence of polynomials* (P_n) *so that*

$$\lim_{n\to\infty} P_n(z) = \begin{cases} 1, & \textit{if } \operatorname{Re}(z) > 0, \\ 0, & \textit{if } \operatorname{Re}(z) = 0, \\ -1, & \textit{if } \operatorname{Re}(z) < 0. \end{cases}$$

Also, explain why (P_n) *can't be bounded on compact subsets of* $\mathbb{C}$.

Solution. For each $n \in \mathbb{N}$, define

$$K_n^+ = \left\{x + iy : \frac{1}{n} \leq x \leq n,\ |y| \leq n\right\},$$

$$K_n^- = \left\{x + iy : -n \leq x \leq -\frac{1}{n},\ |y| \leq n\right\},$$

and

$$K_n^0 = \{iy : |y| \le n\}.$$

Let

$$K_n = K_n^+ \cup K_n^- \cup K_n^0.$$

The set K_n is compact, and its complement is connected. Define $g_n : K_n \to \mathbb{C}$ by

$$g_n(z) = \begin{cases} 1, & z \in K_n^+, \\ -1, & z \in K_n^-, \\ 0, & z \in K_n^0. \end{cases}$$

The three sets K_n^+, K_n^-, and K_n^0 are pairwise disjoint, so g_n is continuous on K_n. Moreover, g_n is holomorphic on the interior of K_n.

By Mergelyan's theorem, there exists a polynomial P_n such that

$$\sup_{z \in K_n} |P_n(z) - g_n(z)| < \frac{1}{n}.$$

Let $z = x + iy \in \mathbb{C}$. If $x > 0$, then for all sufficiently large n,

$$\frac{1}{n} \le x \le n \qquad \text{and} \qquad |y| \le n.$$

Thus $z \in K_n^+$, and hence

$$|P_n(z) - 1| < \frac{1}{n}.$$

Therefore,

$$\lim_{n \to \infty} P_n(z) = 1.$$

If $x < 0$, then for all sufficiently large n, we have $z \in K_n^-$, so

$$|P_n(z) + 1| < \frac{1}{n}.$$

Therefore,

$$\lim_{n \to \infty} P_n(z) = -1.$$

Finally, if $x = 0$, then $z = iy$, and for all sufficiently large n, we have $z \in K_n^0$. Hence

$$|P_n(z)| < \frac{1}{n},$$

and therefore

$$\lim_{n \to \infty} P_n(z) = 0.$$

Consequently,

$$\lim_{n\to\infty} P_n(z) = \begin{cases} 1, & \mathrm{Re}(z) > 0, \\ 0, & \mathrm{Re}(z) = 0, \\ -1, & \mathrm{Re}(z) < 0. \end{cases}$$

It remains to show that (P_n) cannot be bounded on compact subsets of $\mathbb{C}$. Suppose, for contradiction, that it were bounded on compact subsets. Thus, for every compact set $K \subseteq \mathbb{C}$, there would exist a constant M_K such that

$$|P_n(z)| \le M_K \qquad \text{for every } z \in K \text{ and every } n.$$

The family $\{P_n\}$ would then be locally bounded. By Montel's theorem, there would exist a subsequence (P_{n_k}) converging uniformly on compact subsets of $\mathbb{C}$ to an entire function P.

On the other hand, every subsequence of (P_n) has the same pointwise limit

$$G(z) = \begin{cases} 1, & \mathrm{Re}(z) > 0, \\ 0, & \mathrm{Re}(z) = 0, \\ -1, & \mathrm{Re}(z) < 0. \end{cases}$$

Therefore, $P = G$ on $\mathbb{C}$. This is impossible because P is entire and hence continuous, whereas G is discontinuous at every point of the imaginary axis.

Thus, (P_n) cannot be bounded on compact subsets of $\mathbb{C}$. ■

10. *Let $\mathfrak{F}$ be the collection of holomorphic functions on the open unit disk*

$$D = \{z : |z| < 1\}$$

satisfying

$$\iint_D |f|\, dx\, dy \le 1.$$

Prove or disprove that $\mathfrak{F}$ is a normal family.

August 2020

Part I

1. *Let f be an entire function. Show that f is a constant function if there exist $M > 0$ and $R > 0$ such that*

$$|f(z)| \leq M|z|^{1/2} \qquad \textit{whenever } |z| > R.$$

Solution. Expand f in its Taylor series about 0:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

Let $r > R$. By Cauchy's estimate,

$$|a_n| \leq \frac{\max_{|z|=r} |f(z)|}{r^n}.$$

Since

$$|f(z)| \leq M|z|^{1/2}$$

whenever $|z| > R$, we have

$$\max_{|z|=r} |f(z)| \leq Mr^{1/2}.$$

Therefore,

$$|a_n| \leq Mr^{1/2-n}.$$

For every $n \geq 1$, we have $1/2 - n < 0$. Letting $r \to \infty$, we obtain

$$|a_n| = 0.$$

Thus $a_n = 0$ for every $n \geq 1$. Hence

$$f(z) = a_0$$

for all $z \in \mathbb{C}$, and therefore f is constant. ■

2. *Fix $w = re^{i\theta} \neq 0$ and let γ be a piecewise smooth curve in $\mathbb{C} \setminus \{0\}$ from 1 to w.*

Show that there is an integer k such that

$$\int_\gamma \frac{1}{z}\,dz = \log r + i\theta + 2\pi i k.$$

Solution. Let σ be the path from 1 to r along the positive real axis, followed by the circular arc from r to $w = re^{i\theta}$. More precisely, the first part is given by

$$\sigma_1(t) = t, \qquad 1 \le t \le r,$$

and the second part is given by

$$\sigma_2(t) = re^{it}, \qquad 0 \le t \le \theta.$$

Then

$$\int_{\sigma_1} \frac{1}{z}\,dz = \int_1^r \frac{1}{t}\,dt = \log r,$$

and

$$\int_{\sigma_2} \frac{1}{z}\,dz = \int_0^\theta \frac{1}{re^{it}}\,ire^{it}\,dt = i\theta.$$

Therefore,

$$\int_\sigma \frac{1}{z}\,dz = \log r + i\theta.$$

Now let

$$\Gamma = \gamma * (-\sigma),$$

where $-\sigma$ denotes the reverse of σ. Then Γ is a closed curve in $\mathbb{C} \setminus \{0\}$. Hence, by the winding number formula,

$$\int_\Gamma \frac{1}{z}\,dz = 2\pi i k$$

for some $k \in \mathbb{Z}$.

Since

$$\int_\Gamma \frac{1}{z}\,dz = \int_\gamma \frac{1}{z}\,dz - \int_\sigma \frac{1}{z}\,dz,$$

we obtain

$$\int_\gamma \frac{1}{z}\,dz - (\log r + i\theta) = 2\pi i k.$$

Therefore,

$$\int_\gamma \frac{1}{z}\,dz = \log r + i\theta + 2\pi i k$$

for some $k \in \mathbb{Z}$. ■

3. *(a) Let $a > 1$. Show that the equation*

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk $\{z : |z| = 1\}$.

(b) Show that the root in part (a) is positive.

Solution. (a) Define

$$F(z) = ze^{a-z} - 1 \qquad \text{and} \qquad G(z) = ze^{a-z}.$$

On the unit circle $|z| = 1$, we have

$$|G(z)| = |z|e^{a-\operatorname{Re} z} = e^{a-\operatorname{Re} z}.$$

Since $\operatorname{Re} z \le 1$ and $a > 1$,

$$|G(z)| \ge e^{a-1} > 1 = |F(z) - G(z)|.$$

Therefore, by Rouché's theorem, F and G have the same number of zeros in the open unit disk.

Since $e^{a-z} \neq 0$ for every $z \in \mathbb{C}$, the function

$$G(z) = ze^{a-z}$$

has exactly one zero, namely $z = 0$, and this zero is simple. Thus F has exactly one zero in the open unit disk.

Moreover, the strict inequality above shows that F has no zeros on $|z| = 1$. Hence the equation

$$ze^{a-z} = 1$$

has precisely one root in the closed unit disk

$$\{z \in \mathbb{C} : |z| \le 1\}.$$

(b) Consider the real-valued continuous function

$$h(x) = xe^{a-x} - 1, \qquad 0 \le x \le 1.$$

We have

$$h(0) = -1 < 0$$

and

$$h(1) = e^{a-1} - 1 > 0,$$

since $a > 1$. By the intermediate value theorem, there exists $x_0 \in (0,1)$ such that

$$h(x_0) = 0.$$

Thus

$$x_0 e^{a-x_0} = 1.$$

Therefore x_0 is a positive root lying in the unit disk. By the uniqueness established in part (a), it is the unique root in the closed unit disk. Hence that root is positive.

$\blacksquare$

4. *Using the maximum modulus principle, prove the Fundamental Theorem of Algebra.*

Solution. Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0, \qquad a_n \neq 0,$$

be a nonconstant polynomial. We must show that p has a zero in $\mathbb{C}$. Suppose, for contradiction, that $p(z) \neq 0$ for every $z \in \mathbb{C}$. Then

$$f(z) = \frac{1}{p(z)}$$

is an entire function.

Since p is a nonconstant polynomial,

$$|p(z)| \longrightarrow \infty \qquad \text{as} \qquad |z| \longrightarrow \infty.$$

Consequently,

$$|f(z)| = \frac{1}{|p(z)|} \longrightarrow 0 \qquad \text{as} \qquad |z| \longrightarrow \infty.$$

In particular, there exists $R > 0$ such that

$$|f(z)| < |f(0)| \qquad \text{whenever} \qquad |z| > R.$$

Because f is continuous on the compact disk

$$\overline{D(0,R)} = \{z \in \mathbb{C} : |z| \leq R\},$$

the function $|f|$ attains its maximum there. Thus there exists $z_0 \in \overline{D(0,R)}$ such that

$$|f(z_0)| = \max_{|z| \leq R} |f(z)|.$$

Since $0 \in \overline{D(0,R)}$, we have

$$|f(z_0)| \geq |f(0)|.$$

On the other hand, for $|z| > R$,

$$|f(z)| < |f(0)| \leq |f(z_0)|.$$

Therefore, $|f|$ attains its maximum over all of $\mathbb{C}$ at the point z_0. By the maximum modulus principle, f must be constant. Hence $p = 1/f$ is also constant, contradicting the assumption that p is nonconstant. Therefore p must have at least one zero in $\mathbb{C}$. This proves the Fundamental Theorem of Algebra. ∎

5. *Let f be an entire function, taking real values on the real axis, and imaginary values on the imaginary axis. Show that $f(z)$ is an odd function; that is,*

$$f(-z) = -f(z)$$

for all z.

Solution. Since f is entire, it has a Taylor expansion about 0:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n.$$

Because f takes real values on the real axis, we have

$$f(x) \in \mathbb{R} \qquad \text{for all } x \in \mathbb{R}.$$

Therefore all derivatives of f at 0 are real, and hence

$$a_n = \frac{f^{(n)}(0)}{n!} \in \mathbb{R} \qquad \text{for every } n \geq 0.$$

Now let $z = iy$, where $y \in \mathbb{R}$. Then

$$f(iy) = \sum_{n=0}^{\infty} a_n i^n y^n.$$

The real part of this expression is

$$\operatorname{Re} f(iy) = \sum_{k=0}^{\infty} a_{2k}(-1)^k y^{2k}.$$

Since f takes purely imaginary values on the imaginary axis,

$$\operatorname{Re} f(iy) = 0 \qquad \text{for all } y \in \mathbb{R}.$$

Thus

$$\sum_{k=0}^{\infty} a_{2k}(-1)^k y^{2k} = 0 \qquad \text{for all } y \in \mathbb{R}.$$

By uniqueness of power series coefficients,

$$a_{2k} = 0 \qquad \text{for all } k \geq 0.$$

Hence the Taylor series of f contains only odd powers:

$$f(z) = \sum_{k=0}^{\infty} a_{2k+1} z^{2k+1}.$$

Therefore,

$$f(-z) = \sum_{k=0}^{\infty} a_{2k+1}(-z)^{2k+1} = -\sum_{k=0}^{\infty} a_{2k+1} z^{2k+1} = -f(z).$$

Thus f is an odd function. ■

Part II

6. *Given entire functions f, g without common zeros, prove that there exist entire functions A, B such that*

$$Af + Bg = 1.$$

Hint: Use the Mittag-Leffler Theorem.

Solution. Assume first that $f \not\equiv 0$, and let $\{a_n\}$ be the zeros of f, listed without repetition. Since f and g have no common zeros, we have

$$g(a_n) \neq 0$$

for every n.

For each n, let P_n be the principal part at a_n of the Laurent expansion of the meromorphic function

$$\frac{1}{fg}.$$

By the Mittag-Leffler theorem, there exists a meromorphic function h on $\mathbb{C}$ whose only possible poles are the points a_n, and whose principal part at a_n is P_n.

Define

$$B = fh$$

and

$$A = \frac{1}{f} - gh.$$

We claim that A and B are entire. Suppose that a_n is a zero of f of multiplicity m. Since $g(a_n) \neq 0$, the function $1/(fg)$ has a pole of order m at a_n. Hence h has a pole of order at most m at a_n, so the singularity of fh at a_n is removable. Therefore, $B = fh$ is entire.

Moreover, h and $1/(fg)$ have the same principal part at each a_n. Thus

$$\frac{1}{fg} - h$$

is holomorphic at every zero of f. Since

$$A = \frac{1}{f} - gh = g\left(\frac{1}{fg} - h\right),$$

the function A is holomorphic at every zero of f. Away from the zeros of f, it is clearly holomorphic. Hence A is entire.

Finally,

$$Af + Bg = \left(\frac{1}{f} - gh\right) f + (fh)g = 1 - fgh + fgh = 1.$$

Therefore, there exist entire functions A and B such that

$$Af + Bg = 1.$$

If $f \equiv 0$, then g has no zeros, so $1/g$ is entire. In that case, take

$$A = 0, \qquad B = \frac{1}{g}.$$

■

7. *If f is a nonconstant entire function, show that*

$$(f^3 + f^2)(\mathbb{C}) = \mathbb{C}.$$

8. *Suppose that G is a region in $\mathbb{C}$, $\{f_n\}$ is a sequence in $H(G)$, f is a nonconstant function, and $f_n \to f$ in $H(G)$. Let $a \in G$ and $b = f(a)$. Show that there is a sequence $\{a_n\}$ in G such that*

(i) $a = \lim a_n$,

(ii) $f_n(a_n) = b$ *for sufficiently large n.*

9. *Let*

$$G = \{z : \mathrm{Re}(z) > 0\},$$

and $f : G \to G$ *be a one-one analytic function from* G *into* G. *Suppose* $f(a) = a$ *for some positive real number* a. *Show that*

$$|f'(a)| \leq 1.$$

10. *Let* $0 < R_1 < R_2$ *and* $\alpha \in \mathbb{R}$, *and let* G *be the annulus centered at the origin with radii* R_1 *and* R_2. *Let*

$$z^\alpha = \exp(\alpha \log z),$$

where $\log z$ *is a branch of the logarithm. Prove that* z^α *is analytic in* G *if and only if* α *is an integer.*

August 2024

Part I

1. *Let G and Ω be two regions in $\mathbb{C}$, and $h : G \to \mathbb{C}$ be a one-one analytic function, and $f : G \to \mathbb{C}$ be a continuous function with $f(G) \subset \Omega$, and $g : \Omega \to \mathbb{C}$ be an analytic function with $g'(w) \neq 0$ for any $w \in \Omega$. Show that f is analytic if*

$$h(z) = g(f(z)), \qquad \text{for all } z \in G.$$

2. *Let D be the open unit disk. Show that no analytic function $f : D \to \mathbb{C}$ has the property that*

$$|f(z_n)| \to \infty, \qquad \text{whenever } |z_n| \to 1.$$

3. *How many roots does the polynomial*

$$z^9 + z^5 - 8z^3 + 2z + 1$$

have between the circles $\{|z| = 1\}$ and $\{|z| = 2\}$?

4. *Prove Liouville's Theorem by considering the following integral*

$$\int_{C_R} \frac{f(z)}{(z-a)(z-b)}\, dz,$$

for a bounded entire function $f(z)$, where $C_R = Re^{it}$ with $t \in [0, 2\pi]$, and $a \neq b$ with $|a| < R$ and $|b| < R$.

5. *Let G be a region, and*

$$G^* = \{z : \overline{z} \in G\}.$$

If $f : G \to \mathbb{C}$ is analytic, prove that $f^* : G^* \to \mathbb{C}$ defined by

$$f^*(z) = \overline{f(\overline{z})}$$

is also analytic.

Part II

6. *Prove that a nonconstant harmonic function on a region is an open map.*

7. *Let G be a region and $f, f_1, f_2, \ldots$ be elements of $H(G)$. Show that $f_n \to f$ in $H(G)$ if and only if for each closed rectifiable curve γ in G, $f_n \to f$ uniformly on γ.*

8. *Let G be a region and suppose $\mathcal{F}$ is normal in $H(G)$ and Ω is open in $\mathbb{C}$ such that $f(G) \subset \Omega$ for every $f \in \mathcal{F}$. Show that if g is analytic on Ω and is bounded on bounded sets then*

$$\{g \circ f : f \in \mathcal{F}\}$$

is normal.

9. *Show there is an analytic function f on*

$$D = \{z : |z| < 1\}$$

which is not analytic on any region G which properly contains D.

Solution. Let

$$D = \{z \in \mathbb{C} : |z| < 1\},$$

and define

$$a_j = \left(1 - \frac{1}{j}\right) e^{ij}, \qquad m_j = 1, \qquad j \geq 1.$$

Since

$$|a_j| = 1 - \frac{1}{j} < 1,$$

we have $a_j \in D$. The points a_j are distinct because their moduli are distinct. Moreover,

$$|a_j| \longrightarrow 1,$$

so the sequence $\{a_j\}$ has no limit point in D. By Theorem 5.15, there exists an analytic function f on D whose only zeros are the points a_j, each with multiplicity 1. We claim that f cannot be extended analytically to any region properly containing D. Suppose, toward a contradiction, that there exists a region G such that

$$D \subsetneq G$$

and an analytic function F on G satisfying

$$F|_D = f.$$

Since G is open and properly contains D, it contains a point w such that

$$|w| > 1.$$

Because G is a region, it is path connected. Hence there exists a continuous path

$$\gamma : [0,1] \longrightarrow G$$

such that

$$\gamma(0) = 0 \qquad \text{and} \qquad \gamma(1) = w.$$

Since

$$|\gamma(0)| < 1 < |\gamma(1)|,$$

the intermediate value theorem implies that there exists $t_0 \in (0,1)$ such that

$$|\gamma(t_0)| = 1.$$

Set

$$\xi = \gamma(t_0).$$

Then

$$\xi \in G \cap \partial D.$$

Since $\frac{1}{2\pi}$ is irrational, the sequence

$$\{e^{ij}\}_{j=1}^{\infty}$$

is dense in the unit circle. In fact, every tail of this sequence is dense. Therefore, there exists an increasing sequence of integers $\{j_k\}$ such that

$$e^{ij_k} \longrightarrow \xi.$$

Since $j_k \to \infty$, we obtain

$$a_{j_k} = \left(1 - \frac{1}{j_k}\right) e^{ij_k} \longrightarrow \xi.$$

Furthermore,

$$F(a_{j_k}) = f(a_{j_k}) = 0$$

for every k. Thus the zeros of F have a limit point $\xi \in G$. By the identity theorem,

$$F \equiv 0$$

on G. Consequently,

$$f \equiv 0$$

on D, which is impossible because the only zeros of f are the points a_j. Therefore, f is analytic on D and cannot be extended analytically to any region properly containing D. ■

10. *Find an entire function f such that for every integer $n \geq 1$, f has n as a zero of order n. Find the most elementary solution possible.*

January 2025

Part I

1. *Prove the Fundamental Theorem of Algebra by completing the following two steps for a nonconstant polynomial*

$$p(z) = a_0 + a_1 z + \cdots + a_n z^n,$$

where $a_n \neq 0$ and $n \geq 1$.

(a) *Show that there is a nonzero polynomial $q(z)$ such that*

$$\frac{1}{zp(z)} = \frac{1}{zp(0)} + \frac{q(z)}{p(z)},$$

for $z \neq 0$ and $p(z) \neq 0$.

(b) *By means of integration over the circle $C_R = Re^{it}$ with $t \in [0, 2\pi]$, show that $p(z_0) = 0$ for some point z_0 in $\mathbb{C}$.*

2. *Show that the equation*

$$e^z = cz,$$

where c is a constant with $|c| > e$, has exactly one solution in the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

Solution. Let

$$F(z) = e^z - cz.$$

Suppose that z is a solution of

$$e^z = cz$$

with $\operatorname{Re} z < 1$. Taking absolute values, we obtain

$$e^{\operatorname{Re} z} = |c||z|.$$

Therefore,

$$|z| = \frac{e^{\operatorname{Re} z}}{|c|} < \frac{e}{|c|} < 1,$$

since $|c| > e$. Thus, every solution satisfying $\operatorname{Re} z < 1$ lies in the unit disk.

On the circle $|z| = 1$, we have

$$|e^z| = e^{\operatorname{Re} z} \le e < |c| = |cz|.$$

Hence, by Rouché's theorem, the functions

$$e^z - cz \qquad \text{and} \qquad -cz$$

have the same number of zeros in the unit disk, counted with multiplicity.

The function $-cz$ has exactly one zero in the unit disk, namely $z = 0$, with multiplicity one. Consequently, $F(z) = e^z - cz$ also has exactly one zero in the unit disk.

Finally, if $|z| < 1$, then

$$\operatorname{Re} z \le |z| < 1.$$

Thus, the unique zero of F in the unit disk belongs to the set

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

Since every solution in this set must lie in the unit disk, the equation

$$e^z = cz$$

has exactly one solution in

$$\{z \in \mathbb{C} : \operatorname{Re} z < 1\}.$$

■

3. *Show that if $f : \mathbb{C} \to \mathbb{C}$ is a continuous function such that f is analytic off $[-1, 1]$, then f is an entire function.*

Solution. We use Morera's theorem. Since f is continuous on $\mathbb{C}$, it suffices to show that

$$\int_{\partial T} f(z)\, dz = 0$$

for every triangle $T \subset \mathbb{C}$.

First, let P be a polygon contained in the closed upper half-plane. For $\varepsilon > 0$, define

$$P_\varepsilon = P + i\varepsilon.$$

Then P_ε lies entirely in the open upper half-plane, where f is analytic. Hence, by

Cauchy's theorem,

$$\int_{\partial P_\varepsilon} f(z)\,dz = 0.$$

Since f is continuous, letting $\varepsilon \to 0^+$ gives

$$\int_{\partial P} f(z)\,dz = 0.$$

Similarly, by translating downward, the same conclusion holds for every polygon contained in the closed lower half-plane.

Now let T be an arbitrary triangle, and define

$$P_+ = T \cap \{z \in \mathbb{C} : \operatorname{Im} z \geq 0\}, \qquad P_- = T \cap \{z \in \mathbb{C} : \operatorname{Im} z \leq 0\}.$$

The sets P_+ and P_- are polygons, possibly empty or degenerate. By the preceding argument,

$$\int_{\partial P_+} f(z)\,dz = 0, \qquad \int_{\partial P_-} f(z)\,dz = 0.$$

When these two integrals are added, the contributions along the portion of the real axis separating P_+ and P_- cancel, since that segment is traversed in opposite directions. Therefore,

$$\int_{\partial T} f(z)\,dz = \int_{\partial P_+} f(z)\,dz + \int_{\partial P_-} f(z)\,dz = 0.$$

Thus the integral of f around every triangle is zero. By Morera's theorem, f is analytic on all of $\mathbb{C}$. Therefore, f is entire. ■

4. *Using Liouville's Theorem, prove the Fundamental Theorem of Algebra.*

Proof. Suppose $p(z) \neq 0$ for all z, and let

$$f(z) = [p(z)]^{-1}.$$

Then f is an entire function. If p is not constant, then

$$\lim_{z\to\infty} p(z) = \infty,$$

and hence

$$\lim_{z\to\infty} f(z) = 0.$$

In particular, there exists $R > 0$ such that

$$|f(z)| < 1 \qquad \text{if } |z| > R.$$

Since f is continuous on $\overline{B}(0;R)$, there exists a constant M such that

$$|f(z)| \leq M \qquad \text{for } |z| \leq R.$$

Hence f is bounded and therefore must be constant by Liouville's theorem. It follows that p must be constant, contradicting our assumption. ■

5. *Let G be a region and $z_0 \in G$. Suppose $f : G \to \mathbb{C}$ is analytic and $f'(z_0) \neq 0$. Show that*

$$\int_\gamma \frac{1}{f(z) - f(z_0)}\, dz = \frac{2\pi i}{f'(z_0)}$$

if $\gamma = \gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$ and $r > 0$ is small enough.

Part II

6. *Let G and Ω be domains in the complex plane. If $f : G \to \Omega$ is one-to-one and analytic and $\varphi : \Omega \to \mathbb{R}$ is subharmonic, show that $\varphi \circ f$ is subharmonic.*

7. *Let G be a domain in the complex plane and $u : G \to \mathbb{R}$ be a nonconstant harmonic function and let*

$$A = \{z \in G : u_x(z) = u_y(z) = 0\}.$$

Show that A does not have any limit points in G.

Solution. Define

$$f(z) = u_x(z) - i\, u_y(z).$$

Since u is harmonic on G, we have

$$u_{xx} + u_{yy} = 0.$$

Also, harmonic functions are smooth, so

$$u_{xy} = u_{yx}.$$

Thus, the real and imaginary parts of f, namely u_x and $-u_y$, satisfy the Cauchy–Riemann equations:

$$(u_x)_x = (-u_y)_y$$

and

$$(u_x)_y = -(-u_y)_x.$$

Therefore, f is holomorphic on G. Moreover,

$$A = \{z \in G : u_x(z) = u_y(z) = 0\} = \{z \in G : f(z) = 0\}.$$

Hence A is the zero set of the holomorphic function f. Suppose that A has a limit point in G. By the identity theorem,

$$f \equiv 0$$

on G. Consequently,

$$u_x \equiv 0 \qquad \text{and} \qquad u_y \equiv 0$$

on G. Since G is connected, this implies that u is constant on G, contradicting the assumption that u is nonconstant. Therefore, A has no limit points in G. ■

8. *Find an entire function f such that*

$$f\left(ne^{2k\pi i/n}\right) = 0$$

for all positive integers n and all integers $0 \le k \le n-1$. Give the most elementary example possible.

Solution. For each positive integer n, the points

$$ne^{2k\pi i/n}, \qquad 0 \le k \le n-1,$$

are precisely the roots of the polynomial $z^n - n^n$. Thus, consider

$$f(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^n}{n^n}\right).$$

We first show that this product defines an entire function. Fix $R > 0$. For $|z| \le R$, we have

$$\left|\frac{z^n}{n^n}\right| \le \left(\frac{R}{n}\right)^n.$$

If $n \ge 2R$, then

$$\left(\frac{R}{n}\right)^n \le 2^{-n}.$$

Consequently,

$$\sum_{n=1}^{\infty} \sup_{|z| \le R} \left|\frac{z^n}{n^n}\right| \le \sum_{n=1}^{\infty} \left(\frac{R}{n}\right)^n < \infty.$$

Therefore, the infinite product

$$\prod_{n=1}^{\infty}\left(1-\frac{z^n}{n^n}\right)$$

converges uniformly on every compact subset of $\mathbb{C}$. Since its partial products are polynomials, f is entire. Moreover, $f(0)=1$, so f is not identically zero. Now let n be a positive integer and let $0 \le k \le n-1$. Then

$$\left(ne^{2k\pi i/n}\right)^n = n^n e^{2k\pi i} = n^n.$$

Hence the n-th factor in the product vanishes:

$$1-\frac{\left(ne^{2k\pi i/n}\right)^n}{n^n} = 0.$$

It follows that

$$f\left(ne^{2k\pi i/n}\right) = 0.$$

Thus, an elementary example is

$$f(z) = \prod_{n=1}^{\infty}\left(1-\frac{z^n}{n^n}\right).$$

∎

9. *Let G and Ω be two domains in the complex plane. Suppose $\{f_n\}$ is a sequence in $C(G,\Omega)$ which converges to f and $\{z_n\}$ is a sequence in G which converges to a point z in G. Show*

$$\lim_{n\to\infty} f_n(z_n) = f(z).$$

Proof. Since $z_n \to z$ and $z \in G$, there exists $r>0$ such that

$$K = \overline{D(z,r)} \subset G.$$

Moreover, $z_n \in K$ for all sufficiently large n.

For such n, the triangle inequality gives

$$|f_n(z_n)-f(z)| \le |f_n(z_n)-f(z_n)| + |f(z_n)-f(z)|.$$

Because $f_n \to f$ uniformly on the compact set K,

$$|f_n(z_n)-f(z_n)| \le \sup_{w\in K}|f_n(w)-f(w)| \longrightarrow 0.$$

Also, f is continuous and $z_n \to z$, so

$$|f(z_n) - f(z)| \longrightarrow 0.$$

Therefore,

$$|f_n(z_n) - f(z)| \longrightarrow 0,$$

and hence

$$\lim_{n\to\infty} f_n(z_n) = f(z).$$

■

10. *Show that if $\mathcal{F} \subset H(G)$ is normal then*

$$\mathcal{F}' = \{f' : f \in \mathcal{F}\}$$

is also normal.

Solution. Let $\{g_n\}$ be an arbitrary sequence in $\mathcal{F}'$. For each n, choose $f_n \in \mathcal{F}$ such that

$$g_n = f_n'.$$

Since $\mathcal{F}$ is normal, there exist a subsequence $\{f_{n_k}\}$ and a function $f \in H(G)$ such that

$$f_{n_k} \longrightarrow f$$

uniformly on compact subsets of G. By Theorem $VII.2.1$, locally uniform convergence of analytic functions implies locally uniform convergence of all their derivatives. In particular,

$$f_{n_k}' \longrightarrow f'$$

uniformly on compact subsets of G. Since

$$g_{n_k} = f_{n_k}',$$

we obtain

$$g_{n_k} \longrightarrow f'$$

uniformly on compact subsets of G. Thus every sequence in $\mathcal{F}'$ has a subsequence converging uniformly on compact subsets of G to a function in $H(G)$. Therefore, $\mathcal{F}'$ is normal. ■